

ASSIGNMENT #04

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Names : Matthew L. Daniel B. Shiloh G.

Due Date: Friday, October 10th TA's Mailbox (03:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: The following data on $\frac{PV}{RT}$ for Nitrogen gas were reported by Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952).

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

Calculate at temperature 60 °C and pressure 700 bar

1. The thermal[volume] expansion coefficient in **K⁻¹**
2. The molar enthalpy in **kJ/mol**
3. The molar entropy in **kJ/(kmol K)**

In order to we need to Solve this problem given the data we need to create an approximation of the graphs and extrapolate values at a certain temperature and pressure.

Akima splines are a type of piecewise cubic interpolation developed by Hiroshi Akima to create smooth curves through data points without the unwanted oscillations common in standard cubic splines. They work by calculating local slopes based on nearby data, giving more weight to smoother regions and reducing the influence of sharp changes. Each interval between points is represented by a cubic polynomial whose values and first derivatives match at the endpoints, ensuring continuity of the function and its first derivative. Unlike classical cubic splines, Akima splines provide strong local control, are less prone to overshooting, and handle irregular data more realistically.

in the table provided we are given the values in terms of the compressibility factor $Z = PV/RT$ with respect to different temperatures and pressures as seen below: Because We'll

Z Table:						
		Temperature (K)				
P(atm)	P(bar)	273.15	298.15	323.15	348.15	373.15
0	0	1.000	1.000	1.000	1.000	1.000
1	1.01325	1.000	1.000	1.000	1.000	1.000
3	3.03975	0.999	0.999	1	1	1.001
5	5.06625	0.998	0.999	1	1.001	1.001
10	10.1325	0.996	0.998	1	1.001	1.002
30	30.3975	0.959	0.996	1.001	1.005	1.007
50	50.6625	0.958	0.996	1.004	1.009	1.014
100	101.325	0.984	1.004	1.018	1.027	1.033
200	202.65	1.036	1.057	1.072	1.082	1.089
300	303.975	1.132	1.146	1.154	1.16	1.163
400	405.3	1.256	1.254	1.253	1.25	1.247
600	607.95	1.524	1.495	1.471	1.45	1.432
800	810.6	1.798	1.723	1.697	1.658	1.625
1000	1013.25	2.067	1.988	1.923	1.867	1.819

Figure 1: Compressibility Table.

need this in terms of density(ρ) which we obtained by using the Compressibility Factor Equation:

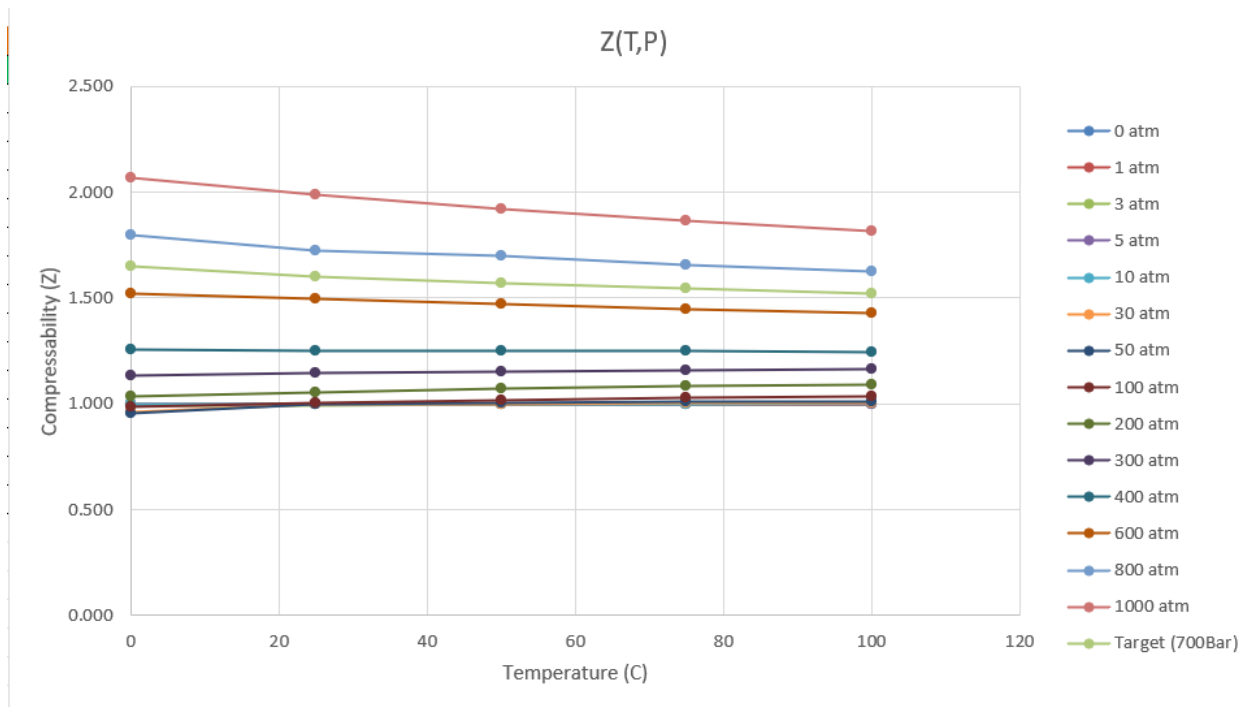
Equation (1)

$$Z = \frac{PV}{RT} = \frac{P}{(\rho)RT}, \quad \rho = \frac{P}{ZRT}$$

To get:

Density Table:						
		Temperature (K)				
P(atm)	P(bar)	273.15	298.15	323.15	348.15	373.15
0	0					
1	1.01325	0.045205	0.041414	0.03821	0.035467	0.03309
3	3.03975	0.13575	0.124367	0.114631	0.1064	0.099172
5	5.06625	0.226477	0.207279	0.191052	0.177155	0.165287
10	10.1325	0.453863	0.414973	0.382103	0.354311	0.330243
30	30.3975	1.414121	1.247419	1.145165	1.058702	0.98581
50	50.6625	2.359328	2.079031	1.902906	1.757509	1.631675
100	101.325	4.593977	4.12493	3.753472	3.453411	3.203327
200	202.65	8.726782	7.836196	7.128795	6.555736	6.077203
300	303.975	11.98005	10.84144	9.933365	9.172378	8.53578
400	405.3	14.39641	13.2103	12.19803	11.34929	10.61439
600	607.95	17.79714	16.62112	15.58546	14.67581	13.86468
800	810.6	20.11334	19.22892	18.01313	17.11292	16.29064
1000	1013.25	21.86973	20.83214	19.87017	18.99653	18.19152

Given the Z values I created a graph to help plot the numbers out and give a visual representation of the compressibility factor with respect to temperature at a constant:



Previously included is the target value of 700 bar at varying temperatures for our spline. This was done via the "fAkima" function provided in the ChemE Calculator provided by Dr. Caracotsios.

the following equation operates as:

$$=fAkima(X_{Data}, F(X)_{data}, \mathbf{Job}, X_{Target})$$

Jobs: 1=Root, 2= Inverse Root, 3= $F'(x)$, 4= $F''(x)$, 5= Integral

Which in this dataset we set our $X_{Data} = P$, $F(X)_{data} = Z$ and our Job = 1 which is the job input used when finding a root of value X_{Target} .

With this in mind we set up a spline function to find Z values at a pressure of 700 bar with varying temperatures(K) and applying **Equation (1)** we get our interpolated density(ρ) values as well:

@ P = 700 bar					
T, K	273.15	298.15	323.15	348.15	373.15
Z	1.648604	1.598531	1.573656304	1.544328501	1.519506
Density, mol/m3	18.93205	17.90274	16.7717445	15.8633729	15.04365
Volume, m3/mol	0.05282	0.055857	0.05962409	0.063038296	0.066473
@ T, K	333.15				
P, bar =	700				
Z	dp/dT)P	p(density)			
1.561807448	-0.03665	16.18015			

Now we can use another "fAkima" function with regards to Temperature and Z values to find a root Z value at $T = 60^\circ C = 333.15K$. The function inputted in is as follows:

$$=fAkima(T_{Data}, Z_{Data}, 1, 333.15)$$

Where we select the T and Z data to find a value for Z at 700bar(Condition previous) and 333.15K which gets us $Z \approx 1.562$. And reapplying **Eq. 1** we get $\rho \approx 16.18 \frac{mol}{m^3}$.

Last identity we need in order to find our Thermal[volume] expansion coefficient(K^{-1}): β . Based on a partial derivation of Enthalpy(H):

(Eq. 2)

$$dH = TdS + VdP$$

$$dH = \left. \frac{\partial H}{\partial T} \right]_P dT + \left. \frac{\partial H}{\partial P} \right]_T dP$$

With identities:

(Eq. 3)

$$C_P = \left. \frac{\partial H}{\partial T} \right]_P dP$$

And 2nd diff. of Enthalpy:

(Eq. 4)

$$\left. \frac{\partial H}{\partial P} \right]_T = T \left(\left. \frac{\partial S}{\partial P} \right]_T \right) + V$$

Combining (Eq.2), (Eq. 3), (Eq. 4)

we get:

$$dH = C_p dT + (T \left. \frac{\partial S}{\partial P} \right]_T + V) dP$$

Combining that with a PDE Maxwell identity we get from Gibb's Equation ($dG = -SdT + VdP$):

We get:

$$\left. \frac{\partial S}{\partial P} \right]_T = - \left. \frac{\partial V}{\partial T} \right]_P$$

which is the preferred format to use because it's in terms of measurable quantities: V, T, and P.

This leads us to another expression of Enthalpy: (Eq. 5.1)

$$dH = C_p dT + (-T \frac{\partial V}{\partial T}]_P + V) dP \implies dH = C_p dT + (-\frac{T}{V} \frac{\partial V}{\partial T}]_P + 1) dP$$

From which we can further simplify the in terms of β : (Eq. 6.1)

$$\beta = \frac{1}{V} \frac{\partial V}{\partial T}]_P$$

Which if you go back to table Our value was in terms of density using the relations between density and volume: $\frac{dV}{V} = -\frac{d\rho}{\rho}$ to get: (Eq. 6.2)

$$\beta = -\frac{1}{\rho} \frac{\partial \rho}{\partial T}]_P$$

With this relation in mind the only thing we have left to do is find the value of $\frac{\partial \rho}{\partial T}]_P$ which we can find by using the "fAkima" function with respect to density and Temperature at constant P:

$$=\text{fAkima}(T_{Data}, \rho_{Data}, \mathbf{3}, 333.15) = \frac{\partial \rho}{\partial T}]_P \approx -0.03665$$

Now we solve for β which is part **A.1**:

$$\beta = -\frac{1}{\rho} \frac{\partial \rho}{\partial T}]_P = -\frac{1}{16.18015} (-0.03665 \frac{1}{mol/m^3} \frac{mol/m^3}{K}) \approx 0.002265 K^{-1}$$

Thermal[volume] expansion coefficient.	
β	0.00226504

Problem A.2 & A.3

Going back to **Eq.5.1**:

$$dH = C_p dT + (-\frac{T}{V} \frac{\partial V}{\partial T}]_P + 1) dP$$

We can fully simplify this in terms of β : (Eq. 5.2)

$$dH = C_p dT + (1 - \beta T) V dP$$

Our next step from here is a little tricky. Now because we have everything in terms of dH , we need to integrate our data to solve for H . However, We don't know the C_P value. When we confronted Dr. Caracatsios about this matter he told us that we can assume that we will hold temperature constant. When asked "why?" he stated that $H^{IG} = C_P dT$ which we can use the function:

$$H^{IG} = C_P dT \implies \{=\text{HIG}(\text{component1}, T(^{\circ}C))\}$$

And calculate the rest under the assumption of Constant Temperature, adding the H^{IG} back in later to solve for total enthalpy dH . This same principle can be applied to Entropy equation:

$$dS = C_P \frac{dT}{T} - \beta V dP$$

$$S^{IG} = C_P \frac{dT}{T} \Rightarrow \{=SIG(\text{component1}, T(^{\circ}C))\}$$

R, Latm/molK					
0.08206					
p @ T = 333.15K	V	dp/dT)P	β	(1- β T)V	- β V
0.037067667	26.97769	-0.00011	0.00300848037	-0.061380612	-0.08116
0.111198577	8.992921	-0.00033	0.00300569627	-0.012119859	-0.02703
0.185253001	5.398023	-0.00057	0.00305090464	-0.088575512	-0.01647
0.370500768	2.69905	-0.00113	0.00304961291	-0.043126876	-0.00823
1.109483869	0.90132	-0.00354	0.00318954229	-0.056419038	-0.00287
1.842899796	0.542623	-0.00595	0.00322619009	-0.040591046	-0.00175
3.627143184	0.275699	-0.01226	0.00338080738	-0.034825198	-0.00093
6.887660237	0.145187	-0.02341	0.00339928437	-0.019233177	-0.00049
9.615077475	0.104003	-0.03105	0.00322978874	-0.007904681	-0.00034
11.84277476	0.08444	-0.0345	0.00291302561	0.002493125	-0.00025
15.20880312	0.065751	-0.03684	0.00242256431	0.012684936	-0.00016
17.63839232	0.056695	-0.03652	0.00207046700	0.017587993	-0.00012
19.51095924	0.051253	-0.03524	0.00180607080	0.020414553	-9.3E-05

So now, in order to simplify our Enthalpy equation we evaluate our data at a constant Temperature($T = 333.15K$) and varying Pressures(P) to we create a new table containing the values in terms of density(ρ), Volume($V = \frac{1}{\rho}$), change in density with respect to Temp. at constant Pressure: $\frac{\partial \rho}{\partial T}]_P$, and β .

$$\rho_{P=x(\text{bar})} = \{=fAkima(T_{Data}, \rho_{Data}, 1, 333.15)\} @ \text{varying Pressures, } P = x(\text{bar})$$

And Same thing for $\frac{\partial \rho}{\partial T}]_P$:

$$\frac{\partial \rho}{\partial T}]_{P=x(\text{bar})} = \{=fAkima(T_{Data}, \rho_{Data}, 3, 333.15)\} @ \text{varying Pressures, } P = x(\text{bar})$$

And because we know that $V = \frac{1}{\rho}$, we can solve for β by using:

(Eq. 6.3)

$$\beta = -V \frac{\partial \rho}{\partial T}]_P$$

With this we get our beta column which we can use to find a value of $(1 - \beta TV)$.

The reason why we solve for $(1 - \beta TV)$ is because that value is constant(C) at that temperature. Essentially having:

$$dH = CdP, \quad C = (1 - \beta T)$$

Now using the fAkima function again to integrate and get our residual enthalpy:

$$\int dH = \int CdP$$

Based on format:

$$=fAkima(P_{Data}, (1 - \beta T)V_{Data}, 5,,Limit_{Lower}, Limit_{Higher})$$

Our input is:

$$=fAkima(P_{Data}, (1 - \beta T)V_{Data}, 5,,(0 \text{ bar}), (700 \text{ bar})) \Rightarrow$$

$$(1 - \beta T)VdP \approx -5.8076 \frac{\text{bar}L}{\text{mol}} \approx -0.0580758 \frac{\text{kJ}}{\text{mol}}$$

We take a similar approach to solve our entropy calculations with with our entropy calculation being:

$$dS = C_P \frac{dT}{T} - \beta V dP$$

Where we solve keeping Temperature constant with changing pressure and then multiplying our previously calculated values to create a column of: $-\beta V$.

And again we use the fAkima integration function on the ChemE Calculator:

$$=fAkima(P_{Data}, -\beta V_{Data}, 5,,(0 \text{ bar}), (700 \text{ bar})) \Rightarrow$$

$$-\beta V dP \approx -0.70944 \frac{\text{bar}L}{\text{mol}K} \approx -70.944 \frac{\text{kJ}}{\text{kmol}K}$$

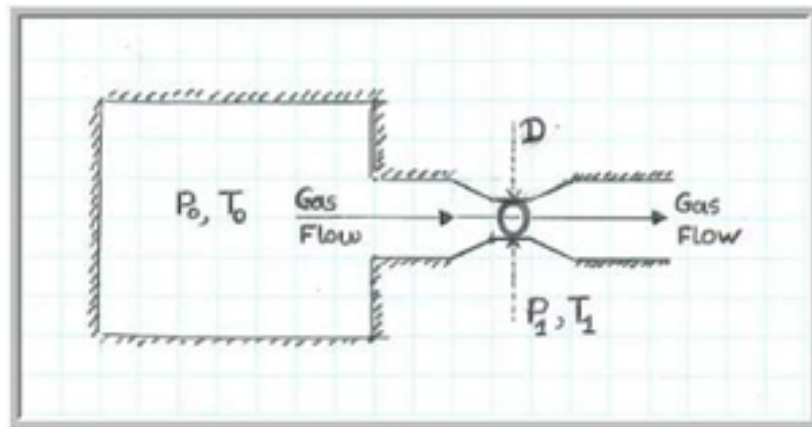
Now all that's left is to calculate the change in enthalpy and entropy. Which is equal to the sum of the Enthalpy Ideal Gas(**HIG(comp1,60°C)**) or Entropy Ideal Gas(**SIG(comp1, 60°C)**) where comp1 = component N_2 . at temperature $60^\circ C$ With our integrated value to get dH and dS:

$$dH = 1.021 \frac{\text{kJ}}{\text{mol}} + (-0.580758 \frac{\text{kJ}}{\text{mol}}) \approx 0.44026 \frac{\text{kJ}}{\text{mol}}$$

$$dS = -51.1213 \frac{\text{kJ}}{\text{kmol}K} + (-70.944 \frac{\text{kJ}}{\text{kmol}K}) \approx -122.065 \frac{\text{kJ}}{\text{kmol}K}$$

Enthalpy Calculations:		Entropy Calculations	
HIG, kJ/kmol	1021.01763	SIG,kJ/kmol-K	-51.1213
H, bar*L/mol	-5.80758004	S barL/mol-K	-0.70944
H kJ/kmol	-580.758004	S, kJ/kmol-K	-70.944
dH, kJ/kmol	440.259622	dS kJ/kmol-K	-122.065

Problem B: Consider the venting of ethylene gas from the carbon steel tank shown in the diagram below



As the pressure begins to drop the gas velocity increases until it reaches the speed of sound in the narrow section of the pipe. **This is known as the critical condition.** The initial pressure and temperature in the tank are equal to 200 bar and 400 K respectively. The volume of the tank is equal to 20 m³

Calculate:

1. The initial mass in **kg** of ethylene in the tank
2. The temperature and pressure at the narrow section of the pipe when the critical mass flow rate which corresponds to the speed of sound (critical condition) is $m=5000$ kg/hr.
3. The diameter in **mm** of the narrow section of the venting pipe
4. The mass in **kg** of ethylene in the tank when the critical condition is reached
5. When during the venting process the pressure gauge reads 2300 **psig** what is the mass in **kg** of ethylene in the tank

Assumptions and Hints

1. Assume that the venting process is **reversible** and **adiabatic**
2. Assume that ethylene behaves as an ideal gas
3. To solve the problem, you must start with the unsteady-state mass, energy and entropy balances
4. The kinetic energy of the gas at the narrow section of the pipe cannot be ignored.

Problem B: Unsteady-State Venting of an Ethylene Tank

Introduction and Problem Statement

This analysis examines the process of venting ethylene gas from a high-pressure storage tank through a restricted pipe section (nozzle). The initial state of the gas in the 20 m³ tank is 200 bar and 400 K. As the gas escapes, it accelerates, and under certain conditions, the flow becomes choked, meaning the velocity at the narrowest point reaches the local speed of sound. This problem requires an analysis of the initial state of the system, the conditions in the throat during choked flow, and the state of the tank after partial depressurization.

The process is governed by the principles of isentropic expansion and unsteady-state mass and energy balances, as instructed by the problem hints. The importance of this study lies in safety and equipment design: engineers must ensure that venting systems prevent catastrophic overpressure, while sizing the vent correctly requires a solid understanding of compressible fluid dynamics.

The safe storage and controlled venting of pressurized gasses such as ethylene is a critical concern in chemical engineering practice. In many industrial settings, ethylene is stored at high pressure in large tanks to minimize volume and facilitate downstream use in polymerization or synthesis. However, high pressure storage introduces hazards and, when rapid depressurization is required, the venting system must relieve pressure safely without causing mechanical failure or uncontrolled releases.

Designing for this condition is challenging because real gases deviate from ideal behavior at such high pressures, expansion causes rapid cooling that can induce condensation or phase change, and unsteady depressurization requires careful tracking of the time dependent tank inventory. In addition, high velocity choked flow can erode nozzle surfaces and cause vibration, while vent sizing must balance cost, operability, and safety margins. Highlighting these factors emphasizes why analyzing venting processes is not only a thermodynamic exercise but also a key step in ensuring safe, reliable chemical plant operation.

In this problem, we also assume that the venting process is both reversible and adiabatic, meaning no heat is exchanged with the surroundings and no entropy is generated inside the system. This idealization implies that the expansion is isentropic, allowing us to apply simplified relations between pressure, temperature, and density at both the tank and throat. While real venting processes often involve some heat transfer to the tank walls and irreversibilities due to turbulence, friction, and shock formation, the reversible-adiabatic assumption provides a clear thermodynamic baseline. It makes the analysis tractable for hand calculations while still capturing the dominant physics of rapid depressurization.

Detailed Analysis: Initial Tank Conditions

Purpose and Function

The first step is to establish the initial thermodynamic state and the total mass (inventory) of ethylene contained within the tank. This initial mass serves as the baseline for all subsequent calculations related to the venting process.

Equations Used

The initial number of moles, n_0 , is determined using the ideal gas law, as specified in the problem assumptions.

$$P_0 V = n_0 R T_0 \quad (1)$$

This assumes ethylene behaves ideally at the initial state, which is an approximation because real-gas effects are present at 200 bar. Still, it provides a helpful starting point. The initial mass, m_0 , is then found by multiplying the number of moles by the molar mass of ethylene ($M_W \approx 28.05 \text{ kg/kmol}$).

$$m_0 = n_0 \times M_W \quad (2)$$

Calculations and Result (Question 1)

Given the initial conditions $P_0 = 200 \text{ bar}$, $T_0 = 400 \text{ K}$, and tank volume $V = 20 \text{ m}^3$:

$$n_0 = \frac{(200 \text{ bar})(20 \text{ m}^3)}{(0.083145 \frac{\text{bar} \cdot \text{m}^3}{\text{kmol} \cdot \text{K}})(400 \text{ K})} = 120.27 \text{ kmol}$$
$$m_0 = 120.27 \text{ kmol} \times 28.05 \frac{\text{kg}}{\text{kmol}} = 3374.0 \text{ kg}$$

Answer: The initial mass of ethylene in the tank is **3374.0 kg**.

Detailed Analysis: Critical (Sonic) Flow at the Throat

Purpose and Function

This section determines the thermodynamic conditions (temperature, pressure) at the narrowest section of the vent pipe (the throat) at the moment the flow becomes critical (choked). This state corresponds to the maximum possible mass flow rate for the given upstream conditions. These calculations are essential for correctly sizing the vent pipe diameter. Choked flow represents a physical limit because no matter how much lower the downstream pressure becomes, the mass flux cannot increase once the throat reaches sonic velocity.

How It Works

As the gas expands from the high-pressure tank (stagnation conditions) into the pipe, its thermal energy is converted into kinetic energy. The process is assumed to be reversible and adiabatic, meaning it is isentropic. For a large enough pressure drop, the velocity at the throat reaches the local speed of sound. At this point, the flow is choked, and the mass flow rate is maximized. The properties at the throat can be calculated directly from the stagnation properties in the tank.

Methods & Equation Used

In order to find Temperature(K) and Pressure inside the neck of the apparatus our group used the Newton Method provided by the ChemE Calculator:

Calculations:	in tank	Neck of pipe	Leak process
R, Latm/kmolK	0.08206	8.31459	
P, bar	200	112.9843063	159.5923908
T, K	400	365.3828599	386.2184694
V, m3	20		
m, kg	3374.012348		
critical Mass Flow, kg/hr		5000	
density, kg/m ³		168.7006174	
v, m/s		359.8847762	
H, kJ/kmol	57323.97357	55507.23788	56586.3139
S, kJ/kmolK	-82.7063037	-82.7063043	-82.7063038
CpIG, kJ/kmol-K	54.20603068	50.73029833	52.84046535
CvIG, kJ/kmol-K	45.89156806	42.41583571	44.52600273
γ	1.181176259	1.196022605	1.186732743
Enthalpy balance, kJ/mol-K	0	0	
Entropy balance, kJ/mol-K	5.60578E-07	0	7.20628E-08
Area, (m ²)/kmol	2.28764E-05		
Diameter, m	0.005396954		
Diameter, mm	5.396954259		
mass @ critical cond., kg	3374.012348		
pressure guage, psig	2300		
pressure guage, psiA	2314.7		
pressure guage, bar	159.5923908		
mass in tank at 2300psig	2788.404697		

Newton Method

Left Hand Side(LHS) Cell:

'Problem B'!\$C\$24:\$C\$25

Solve

Right Hand Side(RHS) Cell:

'Problem B'!\$D\$24:\$D\$25

Cancel

Changing Cell:

'Problem B'!\$D\$8:\$D\$9

☐ Make Unconstrained Variables Non-Negative

Convergence Tolerance:

1.0E-6

ReLoad

Number of Iterations:

30

Clear

This sets up and solves a system of equations in order to get P and T when our enthalpy and entropy balances are equal to each other because the process is reversible, and adiabatic so both enthalpy and entropy are conserved.

The velocity at the throat is the speed of sound:

$$v_1 = \sqrt{\frac{\gamma RT_1}{M_W}} \quad (3)$$

The required vent diameter (D) is calculated from the mass flow rate equation:

$$\dot{m} = \rho_1 A v_1 = \rho_1 \left(\frac{\pi D^2}{4} \right) v_1 \implies D = \sqrt{\frac{4 \dot{m}}{\pi \rho v}} = \sqrt{\frac{4(1.389 \frac{kg}{s})}{\pi (168.7 \frac{kg}{m^3}) (359.88 \frac{m}{s})}}$$

$$\implies D = 0.005397m = 5.397mm$$

Calculations and Results (Questions 2 & 3)

Using the initial tank conditions as the stagnation state and a given mass flow rate of $\dot{m} = 5000 \text{ kg/hr}$ (1.389 kg/s):

- **Throat Temperature and Pressure:** Based on an iterative solution accounting for temperature-dependent heat capacities, the conditions are found to be:
 - $T_1 = 365.4 \text{ K}$
 - $P_1 = 113.0 \text{ bar}$

These results show that while the tank gas is at 400 K and 200 bar, the sonic throat cools significantly (to 365 K) and loses pressure (to 113 bar) as energy is converted into kinetic form.

- **Vent Diameter:** The velocity at the throat is $v_1 = 359.9 \text{ m/s}$. To achieve the specified mass flow, the required diameter is calculated to be $D = 5.40 \text{ mm}$. Such a small throat size illustrates how dense, high-pressure gases can sustain high mass flow even through narrow nozzles

Note: While the problem specifies ideal gas behavior, the provided calculations for the throat density (168.7 kg/m^3) suggest a real gas model was used for higher accuracy, as ethylene deviates significantly from ideality at these pressures. The ideal gas assumption was retained for the in-tank expansion analysis.

Answers:

- The temperature and pressure at the narrow section are **365.4 K** and **113.0 bar**.
- The diameter of the narrow section is **5.40 mm**.

Detailed Analysis: Unsteady-State Depressurization

Purpose and Function

This section analyzes the state of the gas remaining in the tank after a period of venting. Because the process is assumed to be reversible and adiabatic, the expansion of the gas remaining in the tank is isentropic. This matters because operators must know how much gas is left at different pressures. For example, if safety relief systems or downstream units are being designed.

Derivation of the Stagnation Enthalpy Relation

We begin with the general unsteady-flow energy balance for a control volume (the tank), neglecting potential energy and shaft work, and assuming adiabatic flow:

$$\frac{d}{dt}(mu) = -\dot{m}_e \left(h_e + \frac{v_e^2}{2} \right), \quad (4)$$

where

- m = mass of gas in the tank,
- u = specific internal energy of the gas,
- $\dot{m}_e$ = mass flow rate leaving the tank,
- h_e = specific enthalpy of the gas at the exit,
- v_e = velocity of the gas at the exit.

For an ideal gas, $u = c_v T$ and $h = c_p T$. Substituting these relations into Eq. (4) gives

$$\frac{d}{dt}(mc_v T) = -\dot{m}_e \left(c_p T_e + \frac{v_e^2}{2} \right). \quad (5)$$

Now, for a reversible adiabatic (isentropic) flow along a streamline, the *stagnation enthalpy* is conserved:

$$h_0 = h + \frac{v^2}{2} = \text{constant}. \quad (6)$$

Inside the tank, the bulk velocity is negligible ($v_0 \approx 0$), so

$$h_0 = h_{\text{tank}} = c_p T_{\text{tank}}. \quad (7)$$

At the exit, Eq. (6) becomes

$$h_{\text{tank}} = h_e + \frac{v_e^2}{2}. \quad (8)$$

Thus, we obtain the stagnation enthalpy relation:

$$\boxed{h_0 = h_e + \frac{v_e^2}{2}}. \quad (9)$$

Using this relation, the right-hand side of **Eq. (5)** can be written as

$$-\dot{m}_e \left(h_e + \frac{v_e^2}{2} \right) = -\dot{m}_e h_0. \quad (10)$$

Since $h_0 = c_p T_{\text{tank}}$, the energy balance simplifies to

$$\frac{d}{dt}(mc_v T) = -\dot{m}_e c_p T. \quad (11)$$

This equation can then be integrated to relate the mass and temperature of the gas in the tank during the adiabatic discharge process.

Therefore, starting from the general energy balance, we have shown that:

$$\boxed{H_0 = H_1 + \frac{v^2}{2}}, \quad (12)$$

which represents conservation of total (stagnation) enthalpy along a streamline in an adiabatic, reversible flow.

Calculations and Results (Questions 4 & 5)

- **Mass at Critical Condition:** The problem asks for the mass in the tank at the instant the critical flow rate is 5000 kg/hr. Since the vent diameter was calculated based on the initial tank conditions producing this flow rate, the mass in the tank at this instant is the initial mass. In other words, choking occurs immediately, so no significant depletion has occurred yet.
- **Mass at 2300 psig:** The final pressure is $P_f = 2300 \text{ psig} + 14.7 \text{ psi} \approx 2314.7 \text{ psiA}$. Using the ChemE calculator function we can convert psiA to bar by

$$=\text{PSIAtoBAR}(P_{\text{psiA}}) \implies 159.6\text{bar}$$

From here, because our assumption is that ethylene in the gaseous state is stated to behave as an **Ideal Gas**, we can use our ideal gas equation to solve for mass.

$$PV = nRT \implies PV = \frac{m}{M_w}RT \implies m = \frac{M_w PV}{RT}$$
$$m = \frac{M_w PV}{RT} = \frac{(28.054 \frac{\text{g}}{\text{mol}})(159.6\text{bar})(20\text{m}^3)}{(8.314\text{E} - 5 \frac{\text{bar m}^3}{\text{mol K}})(365.383\text{K})} = 2788.40\text{kg}$$

Answers:

- The mass of ethylene in the tank when the critical condition is reached is **3374.0 kg**.
- The mass of ethylene in the tank when the pressure reads 2300 psig is **2788.4 kg**.

Calculations:	in tank	Neck of pipe	Leak process
R, Latm/kmolK	0.08206	8.31459	
P, bar	200	112.9843063	159.5923908
T, K	400	365.3828599	386.2184694
V, m ³	20		
m, kg	3374.012348		
critical Mass Flow, kg/hr		5000	
density, kg/m ³		168.7006174	
v, m/s		359.8847762	
H, kJ/kmol	57323.97357	55507.23788	56586.3139
S, kJ/kmolK	-82.7063037	-82.7063043	-82.7063038
CpIG, kJ/kmol-K	54.20603068	50.73029833	52.84046535
CvIG, kJ/kmol-K	45.89156806	42.41583571	44.52600273
γ	1.181176259	1.196022605	1.186732743
Enthalpy balance, kJ/mol-K	0	0	
Entropy balance, kJ/mol-K	5.60578E-07	0	7.20628E-08
Area, (m ²)/kmol	2.28764E-05		
Diameter, m	0.005396954		
Diameter, mm	5.396954259		
mass @ critical cond., kg	3374.012348		
pressure guage, psig	2300		
pressure guage, psiA	2314.7		
pressure guage, bar	159.5923908		
mass in tank at 2300psig, kg	2788.404697		

Closing Statement and Resources Used:

This analysis was completed using thermodynamic principles from CHE 301, Chapters 3 and 4. The CHE Calculator spreadsheet was used to perform the detailed calculations for stream properties and balances. ChatGpt aided with LaTeX.